



NATIONAL INSTITUTE OF TECHNOLOGY, PATNA
MID SEMESTER EXAMINATION- May 2023
SESSION: 2022-23, Semester II

Program: B. Tech. and B Tech Dual Degree (ECE)
Course Code: PH25101
Full Marks: 30

Department: Physics
Course Name: Engineering Physics
Duration of Examination: 2 hours

Note: Answer all the questions.

1. Answer all the objective type questions given below, of 2 marks each.

[10]

(i) The work done in displacing a charge 2C through 0.5 m on an equipotential surface is
(a) 0 (b) 4 J (c) $2 \times 6.25 \times 10^{18}$ eV (d) None of these

(ii) The vector function $\vec{F} = x^2 y \hat{j} + y^2 x \hat{i} + z^2 \hat{k}$ is,
(a) Solenoidal (b) Irrotational
(c) Neither Solenoidal Nor Irrotational (d) Both Solenoidal and Irrotational

(iii) The divergence of a position vector is,
(a) Zero (b) 1 (c) 2 (d) 3

(iv) The solid angle subtended by a unit spherical surface bounded by limits
 $\theta: 0 \rightarrow \pi/3$ & $\phi: 0 \rightarrow \pi/2$, at its center would be (in steradian)
(a) $\pi/4$ (b) $\pi/2$ (c) $\pi/16$ (d) 4π

(v) Which of the following is zero for any applicable function (scalar or vector wherever applicable)

(a) grad(div) (b) Div(Grad) (c) Curl(Grad) (d) Curl (Curl)

2. A hydrogen atom (Bohr radius = 0.53 Angstrom) is situated between two metal plates, 1 mm apart, which are connected to opposite terminals of 500 Volt battery. What fraction of the atomic radius does the separation distance "d" amount to roughly. The atomic polarizability (α) for hydrogen atom, can be taken from the expression $\frac{\alpha}{4\pi\epsilon_0} = 0.667 \times 10^{-30} \text{ m}^3$. [10]

3. Discuss the case of Uniform and Non uniform polarization in dielectric materials and derive the expression for surface Polarization charge density and Volume Polarization charge density in appropriate cases. [10]

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END SEMESTER EXAMINATION, SESSION: Jan-June 2023

Program: B. Tech. (ECE)
 Course Code: PH25101
 Course Name: Engineering Physics
 Duration of Examination: 3 Hours

Semester: 2
 Department: Physics
 Full Marks: 60

Note: Answer All Questions

1. Explain the Correction introduced by Maxwell in Ampere circuited law for the time varying field and also show that in vector field E and B follow wave equation. [15]

2. Derive the expression for skin depth of an EM wave propagation in a conducting Medium at low frequencies. In a conducting medium, an electric field is given by expression,

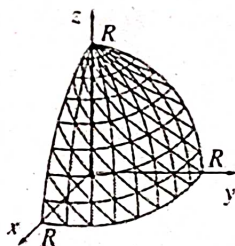
$$\vec{E} = 10e^{-10^8 x} e^{i(4.16 \times 10^7 t - 0.2x)} \hat{z}, \text{ V/m}$$

Find the skin depth of the Electromagnetic wave with its direction (Coefficients have usual SI units) [15]

3. Check the divergence theorem for the function

$$\vec{V} = (r^2 \cos \theta) \hat{r} + (r^2 \cos \phi) \hat{\theta} - (r^2 \cos \theta \sin \phi) \hat{\phi}$$

Using as your volume one octant of the sphere of radius R , as shown in the figure.



Expression for divergence: $\nabla \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$, [10]

4. Explain the construction of Newton's Ring. Show that the diameter of the Brighter rings obtained in reflection mode of Newton's ring experiment, is proportional to the square root of odd numbers. [10]

5. Describe the construction and working principle of Ruby Laser with Schematic diagram of components and Energy levels. [10]